Accelerated Medical Imaging using Machine Learning Architectures

DICOM Integration • Bayesian Uncertainty • Continual Learning

Hospital Network
MRI / X-Ray Machine via DICOM

Mobile App
Doctor's Smartphone via HTTP

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NVIDIA Jetson Nano 4GB (Edge Device)

pynetdicom C-STORE Listener

Flask REST API (:5000)

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STAGE 0: OOD Bouncer
Convolutional Autoencoder
Reconstructs Image → Checks MSE Error

→

UNKNOWN PATHOLOGY
If MSE > Threshold
(Reject & Log)

THE 3-STAGE THREADED PIPELINE (HORIZONTAL PARALLELISM)

STAGE 1: Pre-Process (CPU)
OpenCV Decode → Resize (224x224) → Normalize → CHW format

→ Queue #1 →

STAGE 2: TensorRT (GPU)
Memcpy H2D → FP16 execute_v2() → Extract Logits & 1280-d Maps

→ Queue #2 →

STAGE 3: Post-Process (CPU)
MC Dropout (20x) → True CAM Overlay → Patient Fusion (MLP) → Phi-2 LLM text

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Confidence-Gated Routing
Evaluates Mean Conf% & Std Dev%

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TIER 1 (Auto-Approve)
>90% Conf, <5% Uncertainty
Sends Normal DICOM SR

TIER 2 (Verify)
70-90% Conf OR 5-10% Uncert.
Sends Flagged DICOM SR

TIER 3 (Urgent)
<70% Conf OR >10% Uncert.
Sends Urgent DICOM SR

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CONTINUAL & FEDERATED LEARNING LOOP

1. Active Learning Queue
Stores Tier 3 Uncertain Cases

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2. EWC Fine-Tuning
Loss = task + λ × Σ(Fisher × $\Delta\theta^2$)
Auto Re-compiles TRT Engine

→

3. Federated Server
Averages Weights across 3 Nodes
Hot-swaps back to Pipeline

GPIO Hardware

- Tier 1 (Safe)
- Tier 3 / OOD (Danger)
- Piezo Buzzer (Audio Alert)

LIVE UI DASHBOARD

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Tab 1: Scan Upload
[IMAGE + GradCAM Overlay]
Class: Glioma
Conf: 98.2% ± 1.1%
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Tab 2: Patient Fusion
MRI + XRay + ECG combined.
Unified Risk: CRITICAL
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Tab 3: Active Queue
[Dropdown: Select True Label]
[Button: Submit Correction]
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Tab 4: Tegrastats Monitor
GPU: 99% | CPU: 34%
TEMP: 45°C | PWR: 5.1W
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Tab 5: Federated Sync
Nodes Connected: 3/3
Global Accuracy: 96.4%
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Orthanc PACS Server
Receives all DICOM SR Reports
(External Laptop)

Local SQLite DB
Stores Patient History & Metrics